

Cofunction identities



1 For each of the following acute angles, state the complementary angle:

a 34°

b 62°

c $\frac{\pi}{6}$

d $\frac{2\pi}{7}$

2 Write each of the following trigonometric ratios in terms of its cofunction:

a $\cos 37^\circ$

b $\sin 76^\circ$

c $\tan 68^\circ$

d $\cot(43^\circ)$

e $\operatorname{cosec} 35^\circ$

f $\cos \frac{\pi}{9}$

g $\sin \frac{\pi}{10}$

h $\cot\left(\frac{\pi}{9}\right)$

i $\sec \frac{\pi}{9}$

j $\cos \frac{1}{5}$

3 Consider the following right triangle:

a Find an expression for the following:

i $\cos \theta$

ii $\sin(90^\circ - \theta)$

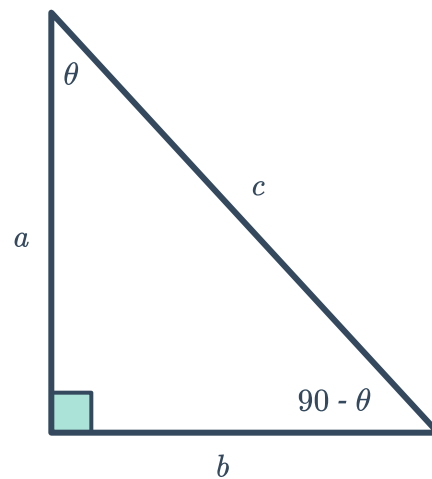
iii $\sin \theta$

iv $\cos(90^\circ - \theta)$

b Describe the rule found in part (a) using words.

c If $\sin \alpha = 0.32$, find $\cos(90^\circ - \alpha)$.

d If $\sin \alpha = \cos \beta$, find $\alpha + \beta$.



4 Simplify:

a $\sin(90^\circ - p)$

b $\frac{\cos(90^\circ - x)}{\cos x}$

c $\frac{\cos 26^\circ}{7 \sin 64^\circ}$

d $\frac{\sin 51^\circ}{\cos 39^\circ}$

e $\sin(90^\circ - y) \times \tan y$

5 Given that $\sin x = 0.19$, find the exact value of $\cos\left(\frac{\pi}{2} - x\right)$.

6 Given that $\sin x = 0.46$, find the exact value of $\cos(90^\circ - x)$.

7 Prove the following:



a $\frac{\sin x \cos (90 - x)}{\cos x \sin (90 - x)} = \tan^2 (x)$

b $\frac{\sin x \sin (90 - x)}{\cos x \cos (90 - x)} = 1$

Trigonometric equations

8 Solve for the measure of the acute angle θ in the following equations:

a $\sin \theta = \cos 25^\circ$

b $\cos \theta = \sin 85^\circ$

c $\cos \theta = \sin \frac{\pi}{8}$

d $\sin \frac{2\pi}{5} = \cos \theta$

9 Solve for the exact value of c , if $\sin \left(\frac{\pi}{9} + c \right) = \cos \left(\frac{\pi}{4} \right)$.

10 Solve for the value of c , if $\sin (34^\circ + c^\circ) = \cos 51^\circ$.

11 Solve for the value of B if $\cos (66^\circ - B) = \sin 38^\circ$.

12 Solve for the value of A if $\sin \left(\frac{A}{68} \right) = \cos 18^\circ$.

13 Solve for the value of x in the following equations:

a $\sin (5x + 40^\circ) = \cos (3x + 10^\circ)$

b $\sin \left(6x + \frac{\pi}{3} \right) = \cos \left(4x + \frac{\pi}{9} \right)$